

Why do geometers care about cones?

Interior regularity of minimal hypersurfaces and Bernstein's theorem

Andrew Lee

Stanford University Research Institute in Mathematics (SURIM)

Summer 2024

Outline

- 1 Plateau's problem
- 2 C^k smooth manifolds revisited
- 3 Generalized surfaces: sets of locally finite perimeter
- 4 Application: Bernstein's theorem

Plateau's problem

Plateau's problem

Suppose we have a compact (smooth) manifold without boundary $\Gamma^{n-1} \subseteq \mathbb{R}^{n+1}$. Consider $\mathcal{S} = \{\text{smooth manifolds } \Sigma^n \subseteq \mathbb{R}^{n+1} \text{ with boundary } \partial\Sigma = \Gamma\}$. Is there a smooth $\Sigma \in \mathcal{S}$ such that $\text{Area}(\Sigma) = \inf_{\Sigma' \in \mathcal{S}} \text{Area}(\Sigma')$?

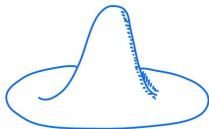
First try: take a sequence $\{\Sigma_i\}_{i=1}^{\infty}$ with area approaching the infimum, and attempt to show convergence.

Plateau's problem

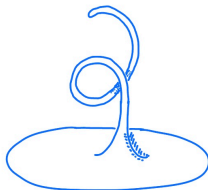
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$$\text{Area} = \pi + 1$$



$$\text{Area} = \pi + \frac{1}{2}$$



$$\text{Area} = \pi + \frac{1}{16}$$

Plateau's problem

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Two problems:

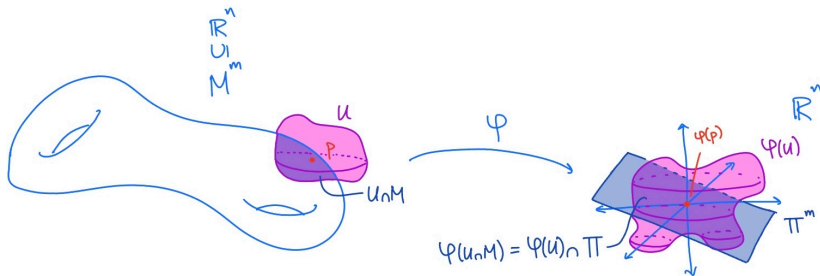
- 1 Need some control on the first derivative to rule out tentacles.
- 2 Need some more technical tools to show that the limit exists and is smooth: define a class of generalized surfaces $\tilde{\mathcal{S}}$ with nice compactness properties.

Motivation: C^k smooth manifolds

Note for experts: For the purposes of my talk, all geometric objects are embedded in a suitably high dimensional Euclidean space. Ask me later if this concerns you!

Definition (C^k smooth submanifold)

A subset $M^m \subseteq \mathbb{R}^n$ is an m -dimensional C^k smooth submanifold if every point $p \in M$ admits a C^k smooth coordinate chart: a pair (U, φ) where $p \in U \subseteq \mathbb{R}^n$ is an open set and $\varphi: U \rightarrow \mathbb{R}^n$ is a C^k diffeomorphism onto its image (i.e., C^k bijection with C^k inverse), such that $\varphi(U \cap M) = \varphi(U) \cap \Pi$, where $\Pi^m \subseteq \mathbb{R}^n$ is a linear m -dimensional subspace.



Motivation: C^k smooth manifolds

Definition (C^k smooth submanifold)

A subset $M^m \subseteq \mathbb{R}^n$ is an m -dim C^k submanifold if $\forall p \in M \exists C^k$ chart: (U, φ) , $p \in U \subseteq \mathbb{R}^n$ open, $\varphi: U \xrightarrow{\sim} \mathbb{R}^n$ is a C^k diffeomorphism onto its image, s.t. $\varphi(U \cap M) = \varphi(U) \cap \Pi$, $\Pi^m \subseteq \mathbb{R}^n$ is a linear m -dim subspace.

Definition* (*tangent space*)

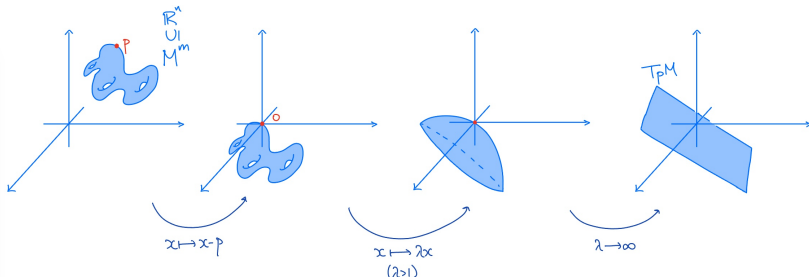
Let $M^m \subseteq \mathbb{R}^n$ be a C^k submanifold, with $k \geq 1$. For every $p \in M$, the *tangent space at p* , denoted $T_p M$, is the linear subspace given by “blowing up at p .”

- 1 Translate p to 0 by mapping $\mathbb{R}^n \rightarrow \mathbb{R}^n$ by $x \mapsto x - p$.
- 2 Dilate (zoom in) by the map $\mathbb{R}^n \rightarrow \mathbb{R}^n$ by $x \mapsto \lambda x$, for $\lambda > 1$.
- 3 Take the limit as $\lambda \rightarrow \infty$.

Motivation: C^k smooth manifolds

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Why does this agree with our usual definition of the tangent space (equivalence classes of curves, directional derivatives, etc.)?

Answer: Assume $p = 0$. Using the implicit function theorem, M is locally a graph $y = f(x)$ for $(x, y) \in \mathbb{R}^m \times \mathbb{R}^{n-m}$, where f is C^k . By Taylor's theorem, $f(x) = df_0(x) + O(|x|^2)$. After the dilation, λM is locally the graph $\lambda^{-1}y = f(\lambda^{-1}x)$, which by Taylor's theorem is expressed as

$$y = \lambda f(\lambda^{-1}x) = \lambda df_0(\lambda^{-1}x) + \lambda O(\lambda^{-2}|x|^2) = df_0(x) + O(\lambda^{-1}|x|^2),$$

so in the limit $\lambda \rightarrow \infty$ we are only left with the linear term.

This proves the $\implies$ direction of:

Claim:

Given a subset $M \subseteq \mathbb{R}^n$ and a point $p \in M$,

$$\exists C^1 \text{ chart } (U, \varphi) \text{ at } p \iff \text{blow-up at } p \text{ is a linear subspace.}$$

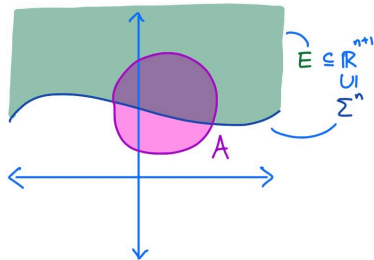
C^1 smooth case

Let $\Sigma^n \subseteq \mathbb{R}^{n+1}$ be a (complete) C^1 submanifold without boundary, and consider the region $E \subseteq \mathbb{R}^{n+1}$ such that $\partial E = \Sigma$.

Let $A \subseteq \mathbb{R}^{n+1}$ be open (or more generally, Borel), and define the *perimeter of E in A* to be $P(E, A) = \text{Area}(\Sigma \cap A)$. Performing a series of (sketchy) manipulations:

$$\begin{aligned} P(E, A) &= \text{Area}(\Sigma \cap A) = \int_{\Sigma=\partial E} \mathbb{1}_A \\ &= \int_E d\mathbb{1}_A = \int_{\mathbb{R}^{n+1}} \mathbb{1}_E d\mathbb{1}_A = - \int_{\mathbb{R}^{n+1}} \mathbb{1}_A d\mathbb{1}_E = - \int_A d\mathbb{1}_E \\ &= \int_A |d\mathbb{1}_E|. \end{aligned}$$

Morally, the perimeter can be thought of as integrating a Dirac delta along the intersection $\Sigma \cap A$.



C^1 smooth case

Let $\Sigma^n \subseteq \mathbb{R}^{n+1}$ be a (complete) C^1 submanifold without boundary, and consider the region $E \subseteq \mathbb{R}^{n+1}$ such that $\partial E = \Sigma$. Let $A \subseteq \mathbb{R}^{n+1}$ be Borel, and define the *perimeter* of E in A by $P(E, A) = \text{Area}(\Sigma \cap A)$.

We can make this precise: since Σ is C^1 smooth, there exists a continuous unit normal $\nu \in C^0(\Sigma \rightarrow \mathbb{R}^{n+1})$, so we can express the perimeter as

$$P(E, A) = \int_{\Sigma} \mathbb{1}_A = \sup \left\{ \int_{\Sigma} \langle g, \nu \rangle : g \in C_c^1(A \rightarrow \mathbb{R}^{n+1}), \sup_{x \in A} |g(x)| \leq 1 \right\}.$$

Using the divergence theorem ($\int_M f \operatorname{div} g = \int_{\partial M} f \langle g, \nu \rangle$) we can equivalently express this by

$$\begin{aligned} P(E, A) &= \sup \left\{ \int_E \operatorname{div} g : g \in C_c^1(A \rightarrow \mathbb{R}^{n+1}), \sup_{x \in A} |g(x)| \leq 1 \right\} \\ &= \sup \left\{ \int_{\mathbb{R}^{n+1}} \mathbb{1}_E \operatorname{div} g : g \in C_c^1(A \rightarrow \mathbb{R}^{n+1}), \sup_{x \in A} |g(x)| \leq 1 \right\}. \end{aligned}$$

Key Observation: if we take the final equality to be the *definition* of $P(E, A)$, we require no smoothness assumption on E .

Functions of bounded variation

Let $\Sigma^n \subseteq \mathbb{R}^{n+1}$ be a (complete) C^1 submanifold without boundary, and consider the region $E \subseteq \mathbb{R}^{n+1}$ such that $\partial E = \Sigma$. Let $A \subseteq \mathbb{R}^{n+1}$ be Borel, and define the *perimeter* of E in A by

$$P(E, A) = \sup \left\{ \int_{\mathbb{R}^{n+1}} \mathbb{1}_E \operatorname{div} g : g \in C_c^1(A \rightarrow \mathbb{R}^{n+1}), \sup_{x \in A} |g(x)| \leq 1 \right\}.$$

Definition (locally bounded variation)

Let $\Omega \subseteq \mathbb{R}^{n+1}$ be open, and let $f \in L_{\text{loc}}^1(\Omega)$ (that is, $\int_A |f| < \infty$ for all $A \subseteq \Omega$ compact). Then, the *variation of f in $A \subseteq \Omega$* is

$$\int_A |Df| = \sup \left\{ \int_A f \operatorname{div} g : g \in C_c^1(A \rightarrow \mathbb{R}^{n+1}), \sup |g| \leq 1 \right\}.$$

If $\int_A |Df| < \infty$ for all $A \subseteq \Omega$ compact, then f is said to be of *locally bounded variation*, denoted $f \in \text{BV}_{\text{loc}}(\Omega)$.

Definition (Caccioppoli sets)

Let $E \subseteq \mathbb{R}^{n+1}$ be a Borel set. Then, E is a *Caccioppoli set*, or a *set of locally finite perimeter*, if $\mathbb{1}_E \in \text{BV}_{\text{loc}}(\mathbb{R}^{n+1})$: $P(E, A) = \int_A |Df| < \infty$ for each $A \subseteq \mathbb{R}^{n+1}$ compact.

Caccioppoli sets

Definition (Caccioppoli sets)

Let $E \subseteq \mathbb{R}^{n+1}$ be a Borel set. Then, E is a *Caccioppoli set*, or a *set of locally finite perimeter*, if $\mathbb{1}_E \in \text{BV}_{\text{loc}}(\mathbb{R}^{n+1})$: $P(E, A) = V(\mathbb{1}_E, A) < \infty$ for each $A \subseteq \mathbb{R}^{n+1}$ compact.

If E has C^1 boundary $\partial E = \Sigma$, then by definition every point $x \in \Sigma$ has a C^1 chart. Amazingly, if E is a Caccioppoli set, this is true almost everywhere: there exists a dense subset $\partial^* E \subseteq \partial E$ called the *reduced boundary*, such that:

- 1 There exists a continuous unit normal $\nu: \partial^* E \rightarrow \mathbb{R}^{n+1}$,
- 2 The blow-up of $\partial^* E$ at each point $p \in \partial^* E$ is a linear subspace,
- 3 $\partial^* E$ is a C^1 hypersurface,
- 4 If $A \subseteq \mathbb{R}^{n+1}$ is open, then $P(E, A) = \text{Area}(\partial^* E \cap A)$.

BV Compactness

Let $f \in \text{BV}_{\text{loc}}(\Omega)$. Define the $W^{1,1}$ norm

$$\|f\|_{W^{1,1}(\Omega)} = \|f\|_{L^1(\Omega)} + \|Df\|_{L^1(\Omega)} = \int_{\Omega} |f| + \int_{\Omega} |Df|.$$

Theorem (BV Compactness)

Let $\{f_i\}_{i=1}^{\infty} \subseteq \text{BV}_{\text{loc}}(\Omega)$ be uniformly bounded on compact sets: that is, $\sup_{i \geq 1} \|f_i\|_{W^{1,1}(A)} < \infty$ for all $A \subseteq \Omega$ compact. Then, there exists a subsequence $\{f_{i_j}\}_{j=1}^{\infty}$ and a function $f \in \text{BV}_{\text{loc}}(\Omega)$ such that $f_{i_j} \rightarrow f$ in $L^1_{\text{loc}}(\Omega)$.

This gives us our desired limit! Recall that in Plateau's problem, we took a sequence $\{\Sigma_i\}_{i=1}^{\infty}$ such that $\text{Area}(\Sigma_i) \rightarrow \inf_{\Sigma' \in \mathcal{S}} \text{Area}(\Sigma')$. Then, let $\{E_i\}_{i=1}^{\infty}$ be Caccioppoli sets corresponding to each Σ_i , and note that the sequence $\{\mathbb{1}_{E_i}\}_{i=1}^{\infty}$ satisfies the hypotheses of BV compactness. Thus there exists a subsequential limit which must be either 0 or 1 a.e., and hence we can assume it is the characteristic function of a Caccioppoli set E . Then, our desired solution is ∂E . But, do we have that ∂E is smooth; i.e., $\partial E = \partial^* E$?

Regularity of minimal Caccioppoli sets

We have found a way to produce a minimizing Caccioppoli set: $E \subseteq \mathbb{R}^{n+1}$ such that

$$P(E, \mathbb{R}^{n+1}) = \text{Area}(\partial^* E) = \inf \{ \text{Area}(\partial^* F) : F \text{ Caccioppoli} \}.$$

Claim*:

Let $E \subseteq \mathbb{R}^{n+1}$ be a minimizing Caccioppoli set. For any $p \in \partial E$, the blow-up of ∂E at p is an area-minimizing cone. Moreover,

- ① $p \in \partial^* E$ if and only if the blow-up is a linear subspace, and
- ② $p \in \partial E \setminus \partial^* E$ if and only if the blow-up is a non-flat area minimizing cone.

Corollary:

Suppose all area-minimizing hypercones in $\mathbb{R}^{n+1}$ are flat. Then, if $E \subseteq \mathbb{R}^{n+1}$ is a minimizing Caccioppoli set, then $\partial E = \partial^* E$. That is, $\Sigma = \partial E$ is smooth.

Simons's Theorem

Theorem (Simons 1968)

Let $E \subseteq \mathbb{R}^{n+1}$ be an area minimizing cone, such that $\partial E \setminus \partial^* E \subseteq \{0\}$. If $n \leq 7$, then ∂E is a hyperplane. If $n \geq 8$, there exist non-flat area-minimizing cones.

Recall that Schoen–Simon–Yau's proof of Bernstein's theorem for $n \leq 5$ relied on the fact that $3 + \sqrt{5} \approx 5.26$. Simons's theorem above relies instead on the fact that $5 + \sqrt{8} \approx 7.82$.

Thus, along with our previous observations, we have a resolution to Plateau's problem for $n + 1 \leq 8$ in codimension 1:

Plateau's problem, $\Gamma^{n-1} \subseteq \mathbb{R}^{n+1}$

Let $n + 1 \leq 8$. For any smooth $\Gamma^{n-1} \subseteq \mathbb{R}^{n+1}$, compact without boundary, there exists a hypersurface $\Sigma^n \subseteq \mathbb{R}^{n+1}$ with $\partial \Sigma = \Gamma$ and

$$\text{Area}(\Sigma) = \inf_{\Sigma' \in \mathcal{S}} \text{Area}(\Sigma').$$

Bernstein's theorem

Recall that a hypersurface $\Sigma^n \subseteq \mathbb{R}^{n+1}$ is called *minimal* if it is a critical point with respect to the area functional. If $\Sigma = \Gamma_u$ is a graph, then Σ is minimal if and only if u satisfies the minimal surface equation (MSE):

$$\operatorname{div} \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right) = 0.$$

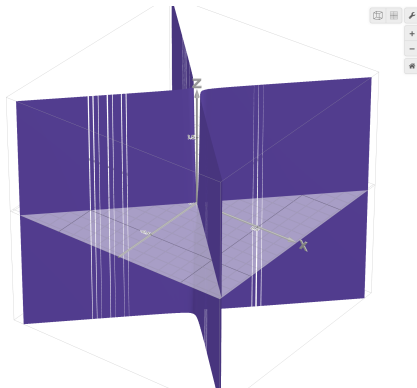
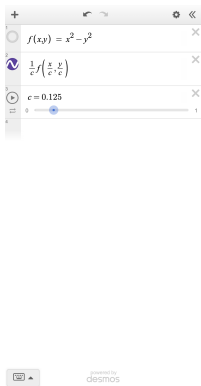
Bernstein's problem

Let $u: \mathbb{R}^n \rightarrow \mathbb{R}$ be an entire solution to the MSE. Then, is Γ_u a hyperplane?

Tangent cone at infinity

Let $\Sigma = \Gamma_u \subseteq \mathbb{R}^{n+1}$. We take the following “blow-down” construction:

- 1 Pick a point $p \in \Sigma$ and translate p to 0.
- 2 Dilate (zoom out) by the map $x \mapsto \lambda x$, for $\lambda < 1$.
- 3 Take the limit as $\lambda \rightarrow 0$.



Tangent cone at infinity

Lemma:

Let $\Sigma = \Gamma_u \subseteq \mathbb{R}^{n+1}$, where u is an entire solution to the MSE. Then,

- ① Σ is area-minimizing.
- ② The tangent cone at infinity $C_\infty \subseteq \mathbb{R}^{n+1}$ is area-minimizing.
- ③ The tangent cone at infinity splits off a line: $C_\infty = C' \times \mathbb{R}$, and $C' \subseteq \mathbb{R}^n$ is an area-minimizing cone.
- ④ If the tangent cone at infinity is a hyperplane, then Γ_u is a hyperplane.

Why does C_∞ split off a line? The blow-down is invariant under translation in the e_{n+1} direction, and hence splits off a line in that direction.

Bernstein's theorem: proof

Bernstein's theorem, $n \leq 7$

Let $u \in C^2(\mathbb{R}^n \rightarrow \mathbb{R})$ be an entire solution to the minimal surface equation, for $n \leq 7$. Then u is affine.

- ① Suppose $n \leq 7$. Then, let $C_\infty \subseteq \mathbb{R}^{n+1}$ be the tangent cone of Γ_u at infinity. By above, C_∞ is area minimizing.
- ② C_∞ splits off a line: $C_\infty = C' \times \mathbb{R}$ for an area-minimizing cone $C' \subseteq \mathbb{R}^n$.
- ③ But, the only area-minimizing cones in $\mathbb{R}^n$ for $n \leq 7$ are hyperplanes, by Simons's theorem.
- ④ Therefore, we must have started from Γ_u a hyperplane.

Thank You!